

Q4.4) 식 (5.43)과 (5.44)를 증명하라.

$$r(A) = n-1 \quad \dots (5.43)$$

$$\mu' A \mu = \beta^2 S_{xx} / \sigma^2 \dots (5.44)$$

$$\begin{aligned} r(A) &= r\left(\frac{I_n - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'}{\sigma^2}\right) = \text{tr}(I_n - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}') \\ &= \text{tr}(I_n) - \text{tr}(\mathbf{1}'\mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}') \\ &= n-1 \end{aligned}$$

$$\begin{aligned} \mu' A \mu &= (\mathbf{x}'\boldsymbol{\beta})' \left(\frac{I_n - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'}{\sigma^2} \right) (\mathbf{x}\boldsymbol{\beta}) = (\mathbf{x}\boldsymbol{\beta})' \left(\frac{I_n - \mathbf{1}\mathbf{1}' + \mathbf{1}\mathbf{1}' - \mathbf{1}\mathbf{1}'}{\sigma^2} \right) (\mathbf{x}\boldsymbol{\beta}) \\ &= (\mathbf{x}_1\boldsymbol{\beta}_1)' \left(\frac{\mathbf{I}_n - \mathbf{1}\mathbf{1}'}{\sigma^2} \right) (\mathbf{x}_1\boldsymbol{\beta}_1) = \beta_1' \sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= S_{xx} \beta_1^2 / \sigma^2 \end{aligned}$$

Q4.6) 단순회귀에서 오차제곱합,

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

을 이차형식 $\mathbf{y}'\mathbf{B}\mathbf{y}$ 로 표현하라. 이 이차형식의 분포를 구하고, 또한 기대치를 $\langle \text{정리 5.1} \rangle$ 을 사용하여 구하라.

$$SSE = \mathbf{y}'(I_n - \mathbf{P}_X)\mathbf{y}, \quad \mathbf{P}_X = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

$$\therefore SSE/\sigma^2 = \mathbf{y}' \left(\frac{I_n - \mathbf{P}_X}{\sigma^2} \right) \mathbf{y} \sim \chi^2(r; \delta), \quad r = \text{rank}(I_n - \mathbf{P}_X) = n-p-1$$

$$\begin{aligned} \delta &= E[\mathbf{y}'\mathbf{A}\mathbf{E}[\mathbf{y}]] = \beta^T \mathbf{X}' \left(\frac{I_n - \mathbf{P}_X}{\sigma^2} \right) \mathbf{X} \beta \\ E[SSE] &= \sigma^2(n-p-1) = \sigma^2(n-p-1) \\ &= \sigma^2(n-2) \quad \therefore p=1 \end{aligned}$$

Q4.5) 단순회귀에서 회귀제곱합,

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

을 이차형식 $\mathbf{y}'\mathbf{A}\mathbf{y}$ 로 표현하라. 이 이차형식의 분포를 구하고, 또한 기대치를 $\langle \text{정리 5.1} \rangle$ 에 의하여 구하라.

$$SSR = \mathbf{y}'(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}')\mathbf{y}$$

$$\begin{aligned} SSR/\sigma^2 &\sim \chi^2(r; \delta), \quad r = \text{rank}(\mathbf{P}_X - \mathbf{P}_1) = \text{trace}(\mathbf{P}_X - \mathbf{P}_1) \\ &= \text{trace}(\mathbf{P}_X) - \text{trace}(\mathbf{P}_1) \\ &= p+1-1 = p \end{aligned}$$

$$\delta = E[\mathbf{y}'\mathbf{A}\mathbf{E}[\mathbf{y}]] = \beta_1' \left(\frac{\mathbf{I}_n - \mathbf{1}\mathbf{1}'}{\sigma^2} \right) \mathbf{X} \beta = \frac{\beta_1' \mathbf{X}_{1,\perp} \mathbf{X}_{1,\perp}' \beta_1}{\sigma^2}$$

$$\mathbf{X}_{1,\perp} = \mathbf{X}_1 - \mathbf{P}_1 \mathbf{X}_1$$

$$\begin{aligned} E[\mathbf{y}'\mathbf{A}\mathbf{y}] &= \sigma^2 E\left[\frac{\mathbf{y}'\mathbf{A}\mathbf{y}}{\sigma^2}\right] = \sigma^2(1+\delta) = \sigma^2 + \beta_1' \mathbf{X}_{1,\perp} \mathbf{X}_{1,\perp}' \beta_1 \\ &= \sigma^2 + \beta_1^2 S_{xx} \end{aligned}$$

$$(\text{or Approach \#2}) : \text{tr}(A\Sigma) + \mu' A \mu = \sigma^2 + \beta_1^2 S_{xx}$$

Q4.7) 식 (5.35)에 있는 총제곱합

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}' \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J} \right) \mathbf{y}$$

의 기대값을 구하고, 앞의 문제 5.5와 5.6의 결과를 사용하여

$$SST = SSR + SSE, \quad E[SST] = E[SSR] + E[SSE]$$

임을 보라.

[Solution]

$$SST = \mathbf{y}' \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J} \right) \mathbf{y}$$

$$SST/\sigma^2 = \mathbf{y}' \left(\frac{\mathbf{I}_n - \mathbf{1}\mathbf{1}'}{\sigma^2} \right) \mathbf{y} \sim \chi^2(n-1, \delta = \beta_1' \mathbf{X}_{1,\perp} \mathbf{X}_{1,\perp}' \beta_1 / \sigma^2)$$

$$\begin{aligned} \therefore E[SST] &= (n-1) + \frac{\beta_1' \mathbf{X}_{1,\perp} \mathbf{X}_{1,\perp}' \beta_1}{\sigma^2} \cdot \sigma^2 \\ &= \sigma^2(n-1) + \beta_1' \mathbf{X}_{1,\perp} \mathbf{X}_{1,\perp}' \beta_1 \\ &= \sigma^2(n-1) + \beta_1^2 S_{xx} \end{aligned}$$

Q4.8) Skip

Q4.9) 단순회귀모형, $y = \beta_0 + \beta_1 x + \epsilon$ 에서 오차항끼리는 서로 독립이고 $N(0, \sigma^2)$ 의 분포를 갖는다고 하자. 오수간의 비(ratio)

$$\theta = -\beta_0/\beta_1$$

에 대한 구간을 다음 절차에 따라서 구하라.

(1) $\bar{y} = b_0 + b_1 \bar{x}$ 이고 $\delta = E[\bar{y}] / E[b_1]$ 라고 놓을 때 다음을 구하라.

$$(a) E[\bar{y} - \delta b_1] \quad (b) \text{Cov}(\bar{y}, b_1) \quad (c) \text{Var}(\bar{y} - \delta b_1)$$

(2) $\bar{y} - \delta b_1$ 의 분포를 구하고, $\text{Var}(\bar{y} - \delta b_1)$ 을 $w\sigma^2$ 으로 놓아라.

(3) σ^2 을 $MSE = S^2$ 로 추정하기로 한다. S^2 과 $\bar{y} - \delta b_1$ 이 서로 독립임을 증명하라.

(4) $t = \frac{\bar{y} - \delta b_1}{S/\sqrt{w}}$ 로 놓으면, 이 통계량은 자유도 $n-2$ 의 t -분포를 하며, 따라서 t^2 은 자유도 $(1, n-2)$ 의 F -분포를 한다.

다음의 부등식

$$t^2 \leq \left[t(n-2; \frac{\alpha}{2}) \right]^2 = F(1, n-2; \alpha)$$

은 δ 의 100(1- α)% 신뢰구간을 갖게 되며, 이 구간을 $d_1 \leq \delta \leq d_2$ 라 할 때 d_1 과 d_2 는 다음의 2차 방정식

$$d^2 \left\{ b_1^2 - \frac{S^2 F(1, n-2; \alpha)}{\sum (x_i - \bar{x})^2} \right\} - 2d \bar{y} b_1 + \left\{ (\bar{y})^2 - \frac{1}{n} S^2 F(1, n-2; \alpha) \right\} = 0$$

의 근이 되어야 함을 보려라.

(5) δ 의 신뢰구간으로부터 $\phi = -\beta_0/\beta_1$ 의 신뢰구간은

$$\bar{x} - d_2 \leq -\beta_0/\beta_1 \leq \bar{x} - d_1$$

임을 보려라.

(Solution)

$$(1) E[\bar{y} - \delta b_1] = E[\bar{y}] - \delta E[b_1] = E[\bar{y}] - \frac{E[\bar{y}]}{E[b_1]} E[b_1] = 0$$

$$(b) \text{Cov}(\bar{y}, b_1) = \text{Cov}\left(\frac{1}{n} \sum_{i=1}^n y_i, \frac{Sxy}{Sxx}\right) = \text{Cov}\left(\frac{1}{n} \sum_{i=1}^n y_i, \sum_{j=1}^n \frac{(x_j - \bar{x})}{Sxx} \cdot y_j\right)$$

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n \frac{(x_j - \bar{x})}{Sxx} \text{Cov}(y_i, y_j) = 0$$

$$(c) \text{Var}(\bar{y} - \delta b_1) = \text{Var}(\bar{y}) + \text{Var}(\delta b_1) - 2\text{Cov}(\bar{y}, \delta b_1)$$

$$= \sigma^2/n + \delta^2 \frac{\sigma^2}{Sxx} - 0$$

$$= \sigma^2 \left(\frac{1}{n} + \frac{\delta^2}{Sxx} \right)$$

$$(2) \bar{y} - \delta b_1 \sim N(0, \sigma^2(n^{-1} + \frac{\delta^2}{Sxx}))$$

$$\stackrel{d}{=} N(0, w\sigma^2)$$

$$(3) S^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n-1} y^T (I_n - \Pi_X) y$$

$$\bar{y} = (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T \mathbf{y}, \quad b_1 = (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{x}^T \mathbf{y}$$

$\therefore \bar{y} \in C(\mathbf{1})$: column space of $\mathbf{1}$

$b_1 \in C(\mathbf{x})$: column space of $\mathbf{x}$

and since $\mathbf{x} = (\mathbf{1} \ x)$, $\bar{y} \in C(\mathbf{1}) \subset C(\mathbf{x}) \therefore \bar{y} \in C(\mathbf{x})$

$$C(I_n - \Pi_X) \perp C(\mathbf{x}) \because \langle (I_n - \Pi_X)a, \mathbf{x}b \rangle = \mathbf{a}^T (I_n - \Pi_X)^T \mathbf{x}b$$

$$= \mathbf{a}^T (\mathbf{x} - \mathbf{x})b = 0$$

$$\therefore \text{AS } \phi = \left(\frac{I_n - \Pi_X}{n-1} \right)^T \mathbf{x} \cdot \mathbf{x} = 0 \Leftrightarrow S^2 \text{ and } (\bar{y} - \delta b_1) \text{ are independent.}$$

$$(4) t = \frac{\bar{y} - \delta b_1}{S/\sqrt{w}} = \frac{\bar{y} - \delta b_1}{\sqrt{w} \sigma} \bigg/ \sqrt{\frac{(n-2)S^2}{\sigma^2}} \bigg/ (n-2) = \frac{Z}{\sqrt{V_1/(n-2)}} \quad \begin{matrix} Z \sim N(0,1) \\ V_1 \sim \chi^2(n-2) \end{matrix}$$

$$\therefore t \sim t(n-2)$$

$$t^2 = \left(\frac{\bar{y} - \delta b_1}{w\sigma^2} \right)^2 \bigg/ \frac{(n-2)S^2}{\sigma^2} \bigg/ (n-2) = \frac{Z^2}{V_1/(n-2)}, \quad Z^2 \sim \chi^2(1), \quad V_1 \sim \chi^2(n-2)$$

$$\therefore t^2 \sim F(1, n-2)$$

$$\text{Confidence interval: } t^2 \leq F_{\alpha}(1, n-2)$$

$$t = \frac{\bar{y} - \delta b_1}{S/\sqrt{w}}, \quad w = (n^{-1} + \frac{\delta^2}{Sxx})$$

$$t^2 S^2 w = (\bar{y} - \delta b_1)^2$$

$$\textcircled{1} t^2 S^2 w = t^2 S^2 \left(n^{-1} + \frac{\delta^2}{Sxx} \right) = t^2 S^2 n^{-1} + \frac{t^2 S^2 \delta^2}{Sxx} = F S^2 n^{-1} + \frac{F S^2 \delta^2}{Sxx}$$

$$\textcircled{2} (\bar{y} - \delta b_1)^2 = (\bar{y} - \delta b_1)^2 = \bar{y}^2 - 2\delta b_1 \bar{y} + \delta^2 b_1^2$$

$$\Rightarrow F S^2 n^{-1} + \frac{F S^2 \delta^2}{Sxx} - \bar{y}^2 - 2\delta b_1 \bar{y} + \delta^2 b_1^2 = 0$$

$$\delta \Rightarrow d : d^2 \left\{ b_1^2 - \frac{S^2 F(1, n-2; \alpha)}{Sxx} \right\} - 2\delta b_1 \bar{y} - \frac{1}{n} S^2 F(1, n-2; \alpha) + (\bar{y})^2 = 0 \quad \square$$

(5) δ 의 신뢰구간으로부터 $\phi = -\beta_0/\beta_1$ 의 신뢰구간은

$$\bar{x} - d_2 \leq -\beta_0/\beta_1 \leq \bar{x} - d_1$$

임을 보려라. (참고문헌 (5.9)에서 발해하였음.)

$$d_1 \leq \delta \leq d_2 \Rightarrow d_1 \leq E[\bar{y}] / E[b_1] \leq d_2$$

$$\Rightarrow d_1 \leq (\beta_0 + \bar{x}\beta_1) / \beta_1 \leq d_2$$

$$\Rightarrow d_1 \leq \beta_0/\beta_1 + \bar{x} \leq d_2$$

$$\Rightarrow -\bar{x} + d_1 \leq \beta_0/\beta_1 \leq -\bar{x} + d_2$$

$$\Rightarrow \bar{x} - d_2 \leq -\beta_0/\beta_1 \leq \bar{x} - d_1 \quad \square$$

5.10) 다음의 방정식

$$y = \frac{a\theta}{a \sin^2 \theta + b \cos^2 \theta}$$

을 어떻게 변환하면 직선회귀의 형태로 바꾸어서 d 와 β 를 최소제곱법에 의하여 추정할 수 있는가를 보려라.

$$y^{-1} = \frac{a \sin^2 \theta + \beta \cos^2 \theta}{a \beta} = \frac{1}{\beta} \sin^2 \theta + \frac{1}{a} \cos^2 \theta$$

$$= \beta^{-1} \sin^2 \theta + a^{-1} \cos^2 \theta$$

Let

- ① $y^{-1} = y'$ ④ $\sin^2 \theta = x_1$
 ② $\beta^{-1} = \beta'$ ⑤ $\cos^2 \theta = x_2$
 ③ $a^{-1} = a'$

$$y' = x_1 \beta' + x_2 a'$$

$$\Pi_{x_1} = x_1 (x_1^T x_1)^{-1} x_1^T, \quad \Pi_{x_2} = x_2 (x_2^T x_2)^{-1} x_2^T$$

$$x_{2,\perp} = x_2 - \Pi_{x_1} x_2, \quad \Pi_{x_{2,\perp}} = x_{2,\perp} (x_{2,\perp}^T x_{2,\perp})^{-1} x_{2,\perp}^T$$

$$\hat{y}' = \Pi_{x'} y' = \Pi_{x_1} y' + \Pi_{x_{2,\perp}} y'$$

$$= x_1 \beta' + \Pi_{x_1} x_2 \hat{a}' + \Pi_{x_{2,\perp}} x_2 \hat{a}'$$

$$\therefore \hat{a}' = (x_{2,\perp}^T x_{2,\perp})^{-1} x_{2,\perp}^T y', \quad \hat{\beta}' = (x_1^T x_1)^{-1} x_1^T y' - (x_1^T x_1)^{-1} x_1^T x_2 \hat{a}'$$

$$\hat{a} = 1/\hat{a}', \quad \hat{\beta} = 1/\hat{\beta}'$$